

IISER KOLKATA
Department of physical sciences
PH4107– Assignment 1
Fresnel equations

Instructor: Prof. Ayan Banerjee

Due Date: August 27, Full marks: 30

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1. (a) (5 points) For the Fresnel equations, prove that the normal component of the electric displacement vector ($\vec{D}$) will be continuous.
 - (b) (3 points) The Fresnel equations have their origin in the propagation of transverse waves along strings having impedances Z_i , where i signifies different strings. The formulae for reflection and transmission of the waves along different strings also remain the same. Given this, explain the sound of the 'swish' of the tail of a whip when it is flashed in air (think about impedance of air compared to that of the whip).
 - (c) (4 points) Explain how a mirage can be formed using your knowledge of Fresnel equations. Use diagrams, and be quantitative wherever required and possible.
 - (d) (4 points) Explain why stars twinkle while planets do not. Be quantitative about your answer, i.e. take into account the representative distances of a star or a planet from the earth, and also consider your eye as a detector and its ability to detect changes in intensity.
 - (e) (3 points) Why will wearing polarized sunglasses help in driving a car during daytime?
 - (f) (6 points) Refer to Fig. 1, and answer the following:
 - (i) What will be the difference between the fringes at detector 1 and 2 ? Explain your answer quantitatively (The experiment is being done in a normal lab, the beam is propagating in the air.) [3]
 - (ii) How do you think the fringes will be modified at the detectors if we insert a linear polarizer in front of BS2 and rotate it? Note that the laser is polarized. Explain clearly. [3].
 - (g) (5 points) Derive an expression for the speed of an evanescent wave in the case of internal reflection in terms of c , n_i and θ_i , where θ_i is the angle of incidence. (i is the angle of incidence)

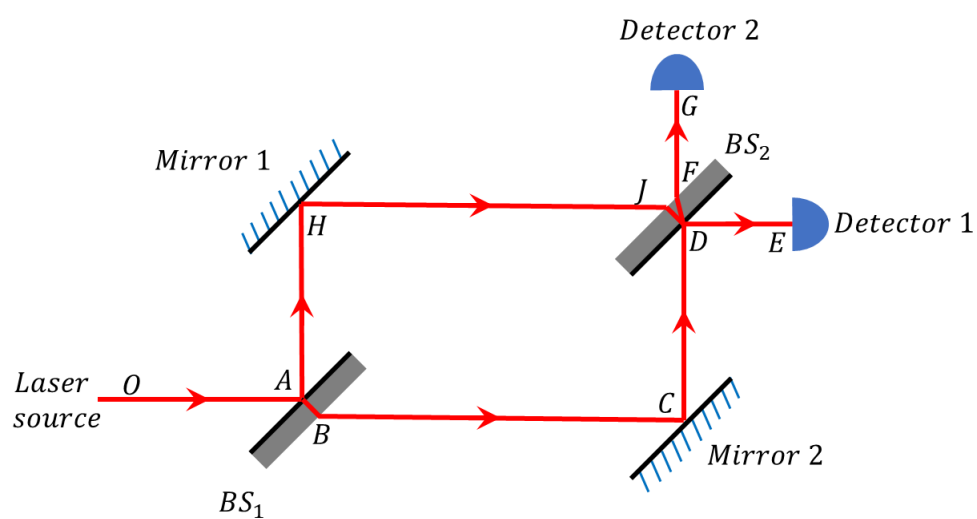


Figure 1: Fig. 1. Mach-Zehnder interferometer